

Improving Front/Back Depth Reasoning in Vision Language Models via Parameter Efficient Spatial Cue Fine Tuning

Jingwu Wang (jingwuwa@andrew.cmu.edu)
Yongyi Xiong (yongyix@andrew.cmu.edu)
Yunxiang Ma (yunxianm@andrew.cmu.edu)

10-423/623 Generative AI Course Project

1 Introduction

Vision language models (VLMs) perform well on general multimodal tasks, yet they persistently struggle with spatial reasoning in partially occluded scenes. To study this bottleneck, we investigate whether lightweight structured spatial cues can improve front/behind depth reasoning. We frame pairwise occlusion order prediction as a generative task using a directionally balanced, image disjoint split of the InstaOrder dataset. Our approach applies parameter efficient instruction tuning with QLoRA to a 7B VLM, Qwen2.5 VL, prompting it with marked images alongside textual spatial metadata, specifically bounding box coordinates and binary 2D overlap flags.

Initial baseline evaluations reveal that the zero shot 7B model exhibits severe directional guessing bias, with 74.53% accuracy on `front` queries and 26.49% accuracy on `behind` queries. Its performance also degrades when objects overlap in 2D. Our study tests whether targeted spatial supervision can mitigate these biases, improve logical pairwise consistency on visually ambiguous pairs, and allow a tuned 7B model to approach or exceed the performance of a larger 72B zero shot reference on this specific task.

2 Dataset and Task

Task. We study *pairwise occlusion order prediction*: given an image and two localized objects A and B , the model decides whether A is in front of or behind B . We formulate this as a generation task, where the VLM answers a natural-language question and the final label is parsed from its response. This gives a unified interface for zero-shot, prompted, and fine-tuned models.

Dataset. We use InstaOrder depth annotations [1] paired with the referenced COCO 2017 images [2]. From the annotations, we extract object pairs, instance masks, bounding boxes, and front/behind labels. After preprocessing, the dataset contains **26,046** train, **2,746**

validation, and **2,952** test samples, with each split balanced at 50.0% `front` and 50.0% `behind`.

Design decisions. Our preprocessing has three main design choices:

- **Directional balance.** Each unordered pair $\{a, b\}$ is emitted twice, as (a, b) and (b, a) , with flipped labels. This prevents a simple `front`-only shortcut from exceeding 50% accuracy.
- **Image-disjoint split.** All pairs from the same COCO image are assigned to only one split, preventing visual leakage across train, validation, and test sets.
- **Marked image variant.** We draw the queried boxes on the image and label them A and B. As shown in Figure 1, swapping A and B produces two directional marked-image variants from the same original image. This makes the queried pair explicit, especially when the same object category appears multiple times.



Figure 1: Dataset example.

Input representations. The baseline and fine-tuning pipelines share the same underlying pairs, splits, labels, and marked-image input. They differ only in whether additional structured textual spatial metadata is provided to the model. Table 1 shows the prompt body variants used for no-cue, bounding-box, overlap, and combined cue settings.

Metrics. We report the evaluation quantities listed in Table 2.

Prompt Variant	Prompt Body Example
No cue	In this image, two objects are highlighted with colored bounding boxes and labels. Object A (marked with a red box) is a person. Object B (marked with a blue box) is a motorcycle. Considering depth and occlusion cues in the scene, is Object A in front of or behind Object B?
BBox	In this image, two objects are highlighted with colored bounding boxes and labels. Object A (marked with a red box) is a person, bounding box [123.0, 45.0, 210.0, 300.0]. Object B (marked with a blue box) is a motorcycle, bounding box [100.0, 80.0, 260.0, 330.0]. Considering depth and occlusion cues in the scene, is Object A in front of or behind Object B?
Overlap	In this image, two objects are highlighted with colored bounding boxes and labels. Object A (marked with a red box) is a person. Object B (marked with a blue box) is a motorcycle. The two bounding boxes overlap. Considering depth and occlusion cues in the scene, is Object A in front of or behind Object B?
BBox+Overlap	In this image, two objects are highlighted with colored bounding boxes and labels. Object A (marked with a red box) is a person, bounding box [123.0, 45.0, 210.0, 300.0]. Object B (marked with a blue box) is a motorcycle, bounding box [100.0, 80.0, 260.0, 330.0]. The two bounding boxes overlap. Considering depth and occlusion cues in the scene, is Object A in front of or behind Object B?

Table 1: Prompt body examples

3 Related Work

Spatial reasoning remains a persistent weakness of VLMs. Kamath et al. [3] show that models that perform strongly on general VQA still fail on basic relations such as left/right and above/below, suggesting that standard pretraining provides weak spatial supervision. This diagnosis is extended by benchmark work. Du et al. [4] evaluate egocentric embodied scenes; Stogiannidis et al. [5] isolate multiple components of spatial reasoning; and Jia et al. [6], Gholami et al. [7] push evaluation toward dynamic, perspective taking, and multiview settings, where the gap between current VLMs and human performance remains large.

A parallel line of work improves spatial reasoning by adding explicit structure. Ranasinghe et al. [8] show that directly supervising image space coordinates improves downstream spatial reasoning. Cheng et al. [9] inject region proposals and depth aware geometric cues, while Fan et al. [10] and Liang et al. [11] extend this direction toward richer 3D representations, including reconstructive 3D tokens and hierarchical 3D supervision. Lee et al. [12] further show that explicit viewpoint transformation can reduce egocentric bias in perspective taking tasks.

Our project is most closely aligned with this shift from implicit to explicit spatial cues, with a narrower focus on occlusion and front/behind reasoning. We do not build a full 3D pipeline. We test whether lightweight structured textual cues can help a VLM resolve a concrete depth ordering relation between two localized objects.

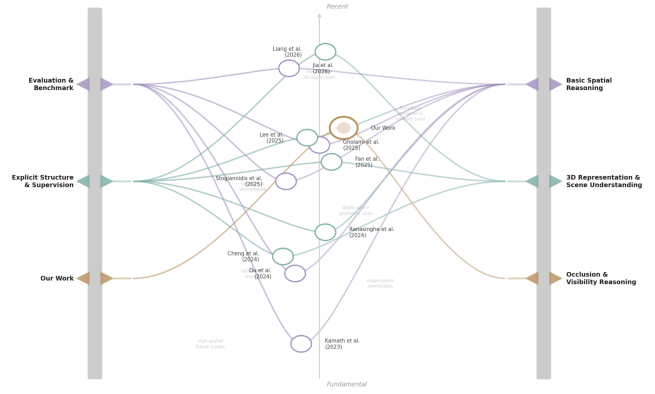


Figure 2: Territory map of related works.

4 Methods

We characterize this as a novel application rather than a new technical method. Importantly, our task is not framed as direct binary classification alone: the model is prompted in a generative vision-language setting, produces a textual response, and the final front/behind label is parsed from that response. All settings use the marked-image input so that object identities are consistently disambiguated across experiments. This lets us isolate the marginal contribution of structured textual cues, rather than conflating cue effectiveness with failures in identifying which instances correspond to object A and object B.

Baseline. We evaluate **Qwen2.5-VL** zero-shot at both 7B and 72B scales on the marked-image test split, using the same prompt template and output parser. The 7B model is the starting point for all fine-tuning experiments, while the 72B model serves as an upper-bound reference for how far pure scaling can take a zero-shot VLM without task-specific supervision.

Zero-shot with spatial cues. Before fine-tuning, we evaluate each cue configuration zero-shot on the 7B model with no parameter updates. This isolates whether the cues themselves are useful to the pre-trained model. That is, whether Qwen2.5-VL can already exploit structured textual coordinates and overlap flags out of the box, given the same image and question. The contrast between these zero-shot cue conditions and their QLoRA counterparts then tells us how much additional benefit comes from learning to use the cues under supervision.

Main method. We fine-tune the 7B baseline with **QLoRA**, which applies low-rank adapters on top of a 4-bit quantized backbone. QLoRA fits our single-GPU compute budget, and because the base weights are frozen, any improvement over the zero-shot baseline is attributable to the adapter and to the structured

Metric	Description
Raw Accuracy	Fraction of test queries answered correctly on the balanced test set.
Accuracy by label	Accuracy computed separately on <i>front</i> and <i>behind</i> queries, which helps detect residual label bias.
Accuracy by overlap	Accuracy split by whether the two queried bounding boxes overlap in 2D, separating pure depth ordering cases from stronger occlusion style cases.
Consistency Score	Fraction of unordered pairs $\{a, b\}$ whose two directional predictions are mutually consistent, meaning one is <i>front</i> and the other is <i>behind</i> .
Pair Accuracy	Fraction of unordered pairs $\{a, b\}$ for which both directional queries are answered correctly.
Front Bias Rate	Fraction of test queries for which the model answers <i>front</i> , regardless of ground truth. This diagnoses prior driven guessing.
Rationale Quality Score	Score assigned by a judge model to each generated rationale, based on whether it is grounded in the image, whether it uses valid depth cues, and whether it contains hallucination.

Table 2: Evaluation metrics used in our experiments.

spatial cues it learns to use. The supervision remains generative: the model is trained to produce a short natural-language answer whose final decision corresponds to *front* or *behind*, rather than to predict a label through a separate classifier head. During training, we optimize the standard next-token cross-entropy loss over the target response.

Varying the textual spatial cue. Holding the backbone, QLoRA configuration, training split, and marked-image input fixed, we vary only the structured spatial information provided in text. We study two such cues: *bbox coordinates* (explicit `bbox_A` and `bbox_B`) and an *overlap signal* (an `overlap` flag indicating whether the queried boxes overlap in 2D). This yields a clean ablation over two kinds of textual spatial metadata: fine-grained instance localization through coordinates, and coarse pairwise interaction through overlap. The resulting configurations are shown in Table 3.

Config	Role	Train	Image	BBox cue	Overlap cue
7B Zero-shot	baseline	—	marked	—	—
7B Zero-shot + bbox	ZS cue	—	marked	yes	—
7B Zero-shot + overlap	ZS cue	—	marked	—	yes
7B Zero-shot + bbox + overlap	ZS cue	—	marked	yes	yes
7B QLoRA + bbox	main	QLoRA	marked	yes	—
7B QLoRA + overlap	main	QLoRA	marked	—	yes
7B QLoRA + bbox + overlap	main	QLoRA	marked	yes	yes
72B Zero-shot	baseline	—	marked	—	—

Table 3: Experimental configurations.

Evaluation protocol. We evaluate the model at two levels. First, for label prediction, we parse the model’s final generated answer into *front* or *behind* and report accuracy-based metrics on the balanced test set. Second, because the model produces free-form textual responses rather than only a hard label, we also evaluate the quality of the generated reasoning. For this part, we mainly use an **LLM-as-a-judge** setup as a signal

for reasoning quality.

What each comparison isolates.

- **7B vs. 72B Zero-shot:** the effect of pure model scale.
- **7B Zero-shot vs. 7B ZS + cue:** whether the pre-trained model can already exploit structured textual cues without any training.
- **7B ZS + cue vs. 7B QLoRA + cue:** the marginal benefit of learning to use the cue under supervision, holding the cue itself fixed.
- **7B QLoRA + bbox vs. 7B QLoRA + overlap:** fine-grained coordinate cues versus coarse overlap cues under identical training.
- **7B QLoRA + bbox + overlap:** whether the two textual cues are complementary when provided together.

5 Experiments

This section reports the final experimental design, results, timing information, and analysis. All experiments use the same filtered, directionally balanced, image disjoint InstaOrder split described in Section 2. Each test query contains a marked image in which the two target boxes are rendered and labeled as *A* and *B*, followed by a pairwise question asking whether *A* is in front of or behind *B*. We compare three groups of settings: 7B and 72B zero shot baselines, three 7B inference only prompt cue variants, and three QLoRA tuned 7B cue variants.

The QLoRA runs use Qwen2.5 VL 7B with 4 bit NF4 quantization, bfloat16 computation, LoRA rank $r = 64$, LoRA scaling $\alpha = 128$, dropout 0.05, learning

Table 4: Results on the filtered InstaOrder test split.

Metric	7B ZS	7B+bbox	7B+ov	7B+bbox+ov	72B ZS	7B ZS+bbox	7B ZS+ov	7B ZS+bbox+ov
Accuracy								
Raw Accuracy	50.51%	92.17%	91.57%	92.65%	70.16%	66.26%	65.18%	66.26%
label=front	74.53%	93.16%	93.36%	93.56%	78.52%	80.28%	86.18%	79.20%
label=behind	26.49%	91.19%	89.77%	91.73%	61.79%	52.24%	44.17%	53.32%
overlap=False	53.02%	95.76%	94.90%	96.08%	74.47%	68.72%	67.41%	68.76%
overlap=True	38.20%	74.60%	75.20%	75.80%	49.00%	54.20%	54.20%	54.00%
bbox.iou								
0 (75.34%)	53.33%	92.72%	92.18%	93.08%	70.64%	66.50%	66.90%	67.57%
(0, 0.1] (14.91%)	45.91%	94.32%	93.18%	93.41%	75.91%	68.51%	63.94%	65.14%
(0.1, 0.3] (7.32%)	38.43%	87.96%	89.81%	91.67%	59.26%	62.50%	53.24%	60.65%
> 0.3 (2.44%)	27.78%	75.00%	68.06%	77.78%	52.78%	56.94%	54.17%	48.61%
Rationale Quality Score (RQS)								
Visual Grounding (1 to 5)	3.91	4.36	4.81	4.80	4.94	4.61	4.47	4.94
Depth Reasoning (1 to 5)	3.25	3.84	4.02	4.03	4.51	4.51	4.39	4.61
Hallucination Rate	6.9%	9.6%	5.4%	3.3%	1.8%	0.0%	0.0%	0.1%
Consistency Score	26.15%	95.87%	95.19%	96.82%	67.01%	53.15%	40.81%	50.63%
Pair Accuracy	21.61%	90.11%	89.16%	91.06%	53.66%	45.34%	40.30%	44.33%
Front Bias Balance	39.94%	98.04%	96.41%	98.17%	83.34%	71.95%	57.95%	74.12%

Note. ZS denotes zero shot prompting. Columns written as **7B+bbox**, **7B+ov**, and **7B+bbox+ov** are QLoRA tuned 7B models. Columns written as **7B ZS+bbox**, **7B ZS+ov**, and **7B ZS+bbox+ov** are inference only prompt cue variants without fine tuning. **bbox** denotes bounding box coordinates, and **ov** denotes the binary 2D overlap cue. Higher values are better except for Hallucination Rate.

Run type	GPU time	Source
7B test inference, one setting	0.60 GPU h	Recorded as $8 \times 4m$ 28s on B200 workers.
72B test inference, one setting	2.55 GPU h	Recorded as $8 \times 19m$ 09s on B200 workers.
7B validation inference, one setting	~ 0.56 GPU h	Estimated by scaling test inference by $2746/2952$.
7B QLoRA training, one cue setting, one epoch	~ 13.2 to 15.9 GPU h	Diary level estimate from 7B inference throughput and forward/backward overhead.

Table 5: Recorded and estimated timing information. The training row is an estimate because the training timer was less cleanly instrumented than the inference runs.

rate 1×10^{-4} , and effective batch size 64. The base model weights remain frozen, so the learned changes are concentrated in the low rank adapters. The main inference jobs were executed on Modal using eight parallel containers, each equipped with one NVIDIA B200 GPU. The 72B zero shot baseline uses the same evaluation pipeline and prompt parser with the larger Qwen2.5 VL model as a scale based reference.

The 7B zero shot model reaches 50.51% raw accuracy, close to the chance level induced by our balanced construction. The aggregate number hides a strong directional shortcut: the model obtains 74.53% accuracy on *front* queries and 26.49% on *behind* queries. Its structural metrics are also weak, with 26.15% Consistency Score and 21.61% Pair Accuracy. Scaling improves the result: the 72B zero shot model reaches 70.16% raw accuracy, 67.01% Consistency Score, and

53.66% Pair Accuracy. However, it still drops from 74.47% on *overlap=False* cases to 49.00% on *overlap=True* cases.

Prompt cue variants provide a useful intermediate result. Adding *bbox*, *overlap*, or both cues to the 7B zero shot prompt raises raw accuracy to roughly 65% to 66%, showing that the pretrained model can partially read the added metadata. Yet pairwise structure remains weak. For example, **7B ZS+ov** reaches 65.18% raw accuracy, while its Consistency Score is only 40.81%. This suggests that textual cues alone do not reliably create a stable relational decision rule.

The QLoRA tuned models produce the central result. All three tuned 7B variants exceed 91% raw accuracy and outperform the 72B zero shot reference on this filtered task. The combined cue setting is strongest overall: **7B+bbox+ov** reaches 92.65% raw accuracy, 93.56% accuracy on *front*, 91.73% on *behind*, 96.82% Consistency Score, and 91.06% Pair Accuracy. These results support our main hypothesis that lightweight structured spatial cues, when paired with parameter efficient tuning, can correct the 7B model’s directional shortcut and produce more stable front/behind judgments.

The main remaining weakness is overlap heavy occlusion. The strongest model, **7B+bbox+ov**, reaches 96.08% on non overlapping examples and 75.80% on overlapping examples. Across **bbox.iou** buckets, it reaches 93.08% when IoU is zero and falls to 77.78% when IoU is greater than 0.3. Thus, the method solves much of the label bias problem and greatly improves pairwise consistency, while deeply ambiguous occlusion cases remain difficult. RQS provides an additional

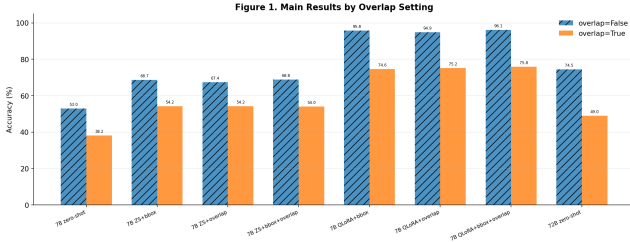


Figure 3: Accuracy under the overlap split.

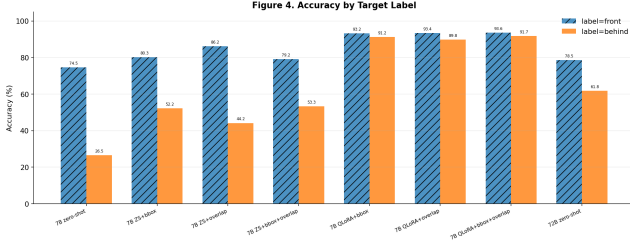


Figure 4: Accuracy by target label.

diagnostic: **7B+bbbox+ov** reaches 4.80 visual grounding and 3.3% hallucination rate, suggesting that its explanations are often grounded, although the strongest evidence of improvement remains the decision level and pairwise metrics.

6 Code Review

Our code review focuses on two core evaluation components: `eval.py` and `judge_rqs.py`. Since our model generates a short rationale and a final answer rather than a direct classification label, these scripts connect generative outputs with both quantitative and qualitative evaluation.

Controlled prompt construction. In `eval.py`, the `build_prompt()` function (lines 137-197) creates different prompt variants for the same front/back reasoning task. The baseline prompt only identifies Object A and Object B, while the other variants add bounding-box coordinates, 2D overlap information, or both. This lets us isolate how different spatial cues affect model performance while keeping the image and question fixed.

Inference and answer parsing. In `eval.py`, lines 304–357 implements batched inference. For each valid sample, `infer_batch()` loads the image, builds the selected prompt variant, applies the processor’s chat template, and generates a 2–3 sentences of spatial reasoning followed by a final answer line in the format `Answer: front` or `Answer: behind`.

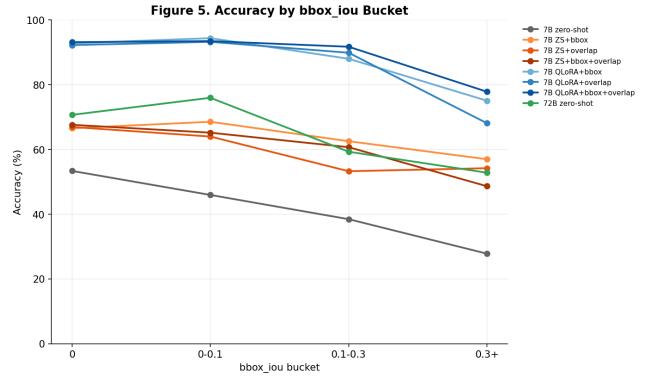


Figure 5: Accuracy across `bbox_iou` buckets.

The `extract_answer()` function (lines 204-234) maps each free-form response to `front`, `behind`, or `unknown`. It first searches for the required format, such as `Answer: front`. Because Qwen2.5-VL is a bilingual vision-language model, we also added fall-back rules for Chinese-style answer markers and Chinese spatial words, in case the model does not fully follow the English-only instruction. This makes the parser more robust to formatting errors and bilingual outputs while still enabling consistent quantitative evaluation.

Rationale quality evaluation. In `judge_rqs.py`, the `JUDGE_SYSTEM_PROMPT` (lines 12-43) defines our Rationale Quality Score evaluation. Instead of only checking whether the final answer is correct, it asks a vision-language judge to evaluate the model’s reasoning along three dimensions: visual grounding, depth reasoning, and hallucination. This helps us test whether a correct answer is supported by real visual evidence and valid depth cues, rather than by shallow label guessing.

Together, `eval.py` evaluates whether the model predicts the correct front/back relation, while `judge_rqs.py` evaluates whether the generated explanation is visually grounded and spatially meaningful.

7 Timeline, Human Work & AI Use

The project work was divided into dataset construction, baseline evaluation, QLoRA tuning, analysis, visualization, and final writing. We used ChatGPT and Claude Code as supporting tools. ChatGPT helped with literature search, wording refinement, and report editing. Claude Code helped generating QLoRA skeleton code and deployment scripts on Modal. Table 6 in Appendix A provides a detailed breakdown of approximate time spent across project stages, human work, AI assistance, and validation procedures.

8 Thought Experiment On Compute

Distributed NVIDIA B200 GPUs on Modal were used for fine-tuning and inference. Fine-tuning each 7B variant required roughly 16 GPU hours. Evaluating the 7B and 72B models consumed approximately 0.60 and 2.55 GPU hours, respectively. Including trials and failed runs, we used about 64 GPU hours in total. At an hourly rate of \$6.25, our total compute cost was approximately \$400.

In our setting, B200 GPUs did not provide a substantial speedup over H200 GPUs. With \$1,000 in AWS credits and a rate of \$6.88 per H200 GPU hour, we would have access to about 145 GPU hours. We would use this additional compute to expand the experiments in three ways. First, we would apply QLoRA directly to the 72B model to test whether explicit spatial cues remain useful at larger scale. Second, we would compare 4-bit QLoRA with stronger adaptation methods for the 7B model. Third, we would run a broader hyperparameter sweep and evaluate additional open-weight VLMs, such as LLaVA-OneVision.

9 Research Log

Our project evolved from a broad interest in VLM spatial reasoning into a controlled study of front/behind occlusion order. At first, we considered many spatial tasks, including left/right relations, embodied navigation, multiview reasoning, and 3D scene understanding. This scope quickly became too broad for a course project because each subproblem required a different dataset and evaluation protocol. InstaOrder allowed us to narrow the question to one concrete relation: given two localized objects in a natural image, can a VLM determine which one is in front? This narrowing gave the project its final form and made the experimental claims measurable.

The most important technical work was building the dataset and evaluation pipeline. InstaOrder annotations are not directly formatted as VLM instruction examples, so we had to extract ordered object pairs, preserve instance identity, render marked images with boxes labeled *A* and *B*, and enforce an image disjoint split. We also emitted each unordered pair in both directions to make the label distribution exactly balanced. This decision became central to the analysis. Because the dataset is balanced, the 7B model’s 50.51% raw accuracy could be read as near chance, while its large gap between `front` and `behind` exposed a directional shortcut.

The first baseline runs changed our evaluation strategy. Raw accuracy alone was too coarse, so we added per label accuracy, overlap splits, `bbox_iou` buckets,

Consistency Score, Pair Accuracy, and Front Bias Balance. The reciprocal pair metrics became especially important because a model can answer isolated examples while still failing to represent the relation as a stable pairwise structure. We then tested prompt only spatial cues to see whether clearer input could solve the problem. These cues improved 7B raw accuracy to about 65% to 66%, but consistency remained weak. This result motivated QLoRA as the practical path forward.

QLoRA made the final experiment computationally feasible. Full fine tuning was outside our budget, and even 72B zero shot evaluation was expensive. With 4 bit NF4 quantization and low rank adapters, we could train 7B cue variants under a controlled setup. The strongest result came from **7B+bbox+overlap**, which greatly reduced the front label shortcut and raised reciprocal consistency. At the same time, the final results clarified the project boundary. The method works very well on this filtered InstaOrder task, yet high overlap cases remain hard, and we have not tested whether this narrow tuning transfers to broader VLM benchmarks.

10 Conclusion

This project tested whether structured spatial cues can improve front/behind depth reasoning in VLMs. The 7B zero shot baseline revealed a severe front label shortcut, with 74.53% accuracy on `front` examples and 26.49% on `behind` examples. The 72B zero shot model improved raw accuracy to 70.16%, yet it still struggled on overlap heavy examples.

Our main finding is that explicit spatial cues become highly effective when paired with QLoRA tuning. Prompt only cues produced moderate gains, while QLoRA tuned cue variants raised raw accuracy above 91%. The strongest configuration, **7B+bbox+overlap**, reached 92.65% raw accuracy, 96.82% Consistency Score, and 91.06% Pair Accuracy, outperforming the 72B zero shot reference on this specific benchmark.

The results have clear limits. Performance still drops on `overlap=True` examples and in high `bbox_iou` buckets, indicating that binary overlap and `bbox` coordinates are insufficient for the hardest occlusion scenes. Future work should evaluate broader spatial benchmarks, check whether task specific tuning harms general VLM capabilities, and explore richer visibility aware cues such as mask overlap, visible area ratio, relative scale, segmentation masks, and depth order priors.

A Detailed Timeline, Human Work, & AI Use

Project Stage	Hours	Human Work	AI Assistance	Validation
Forming research ideas, reading related work, and finding datasets	20	Read papers on VLM spatial reasoning, the InstaOrder dataset, occlusion/depth ordering, and QLoRA-based parameter-efficient fine-tuning.	ChatGPT was used to identify related work and summarize papers.	We only cited papers that we actually read and verified that the datasets were publicly available.
Reading code documentation	10	Studied Qwen2.5-VL, HuggingFace Transformers, QLoRA, Modal.	ChatGPT was used to find relevant APIs and explain their usage.	Verified against official documentation and small test runs.
Understanding existing codebase	5	Inspected InstaOrder dataset annotation format and processing script.	Claude Code helped explain code structure and core files.	Manually reviewed the core files.
Compiling and running existing code	5	Set up the environment, installed dependencies, downloaded referenced COCO/InstaOrder datasets, and ran baseline inference scripts.	Claude Code helped resolving environment issues.	Confirmed that data processing scripts ran end-to-end on small subsets before full runs.
Dataset reconstruction and preprocessing	15	Extracted object pairs, rendered marked images, built balanced front/behind examples, and enforced image-disjoint splits.	ChatGPT helped review the reconstructed datasets.	Checked label balance, image-disjointness, sample counts, and manually inspected rendered examples.
Writing or modifying fine-tuning/evaluation code	25	Wrote and modified prompt builders, answer parsers, metric computation, reciprocal-pair consistency metrics, and RQS evaluation scripts.	Claude Code helped generate initial skeletons for QLoRA, evaluation, and Modal deployment scripts.	Manually reviewed code, ran sanity checks on subset examples, and compared parsed labels with raw model outputs.
Writing scripts to run experiments	5	Built scripts for zero-shot evaluation, prompt-cue evaluation, QLoRA training, distributed Modal inference, and result aggregation.	Claude Code helped draft boilerplate job scripts and Modal deployment logic.	Ran small-scale dry runs before full experiments and checked logs for failed outputs.
Running experiments	30	Launched 7B/72B zero-shot evaluations, 7B prompt-cue evaluations, and three QLoRA fine-tuning variants.	Claude Code were used to debug failed jobs.	Compared logs, GPU-hour records, output counts, and expected dataset sizes.
Compiling results and visualizations	10	Aggregated accuracy, label breakdowns, overlap splits, IoU buckets, consistency score, pair accuracy, and RQS metrics.	ChatGPT helped improve table wording and suggest clearer metric descriptions.	Recomputed metrics from saved outputs and checked consistency between tables and figures.
Presenting the poster	10	Summarized the research into a poster and presented to the class.	ChatGPT voice mode helped practicing the presentation and giving feedback.	Rehearsed based on the feedback.
Writing the report	15	Wrote the problem framing, dataset/task description, methods, experiments, analysis, compute discussion, research log, and conclusion.	ChatGPT was used for grammar and wording refinement.	Proofread the report.

Table 6: Approximate timeline, human work, and AI assistance.

References

- [1] Hyunmin Lee and Jaesik Park. Instance-wise occlusion and depth orders in natural scenes, 2021. URL <https://arxiv.org/abs/2111.14562>.
- [2] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco: Common objects in context, 2014. URL <https://arxiv.org/abs/1405.0312>.
- [3] Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? investigating their struggle with spatial reasoning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9161–9175, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.568. URL <https://aclanthology.org/2023.emnlp-main.568/>.
- [4] Mengfei Du, Binhao Wu, Zejun Li, Xuanjing Huang, and Zhongyu Wei. Embspatial-bench: Benchmarking spatial understanding for embodied tasks with large vision-language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 346–355, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-short.33. URL <https://aclanthology.org/2024.acl-short.33/>.
- [5] Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsafaris. Mind the gap: Benchmarking spatial reasoning in vision-language models, 2025. URL <https://arxiv.org/abs/2503.19707>.
- [6] Mengdi Jia, Zekun Qi, Shaochen Zhang, Wen Yao Zhang, Xinqiang Yu, Jiawei He, He Wang, and Li Yi. Omnispatial: Towards comprehensive spatial reasoning benchmark for vision language models. In *The Thirteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=6nZKT2rL0H>.
- [7] Mohsen Gholami, Ahmad Rezaei, Zhou Weimin, Yong Zhang, and Mohammad Akbari. Spatial reasoning with vision-language models in ego-centric multi-view scenes, 2025. URL <https://arxiv.org/abs/2509.06266>.
- [8] Kanchana Ranasinghe, Satya Narayan Shukla, Omid Poursaeed, Michael S. Ryoo, and Tsung-Yu Lin. Learning to localize objects improves spatial reasoning in visual-llms. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Ranasinghe_Learning_to_Localize_Objects_Improves_Spatial_Reasoning_in_Visual-LLMs_CVPR_2024_paper.html.
- [9] An-Chieh Cheng, Hongxu Yin, Yang Fu, Qiushan Guo, Ruihan Yang, Jan Kautz, Xiaolong Wang, and Sifei Liu. Spatialrgpt: Grounded spatial reasoning in vision-language models. In *Advances in Neural Information Processing Systems*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/f38cb4cf9a5eaa92b3cfa481832719c6-Abstract-Conference.html.
- [10] Zhiwen Fan, Jian Zhang, Renjie Li, Junge Zhang, Runjin Chen, Hezhen Hu, Kevin Wang, Huaizhi Qu, Dilin Wang, Zhicheng Yan, Hongyu Xu, Justin Theiss, Tianlong Chen, Jiachen Li, Zhengzhong Tu, Zhangyang Wang, and Rakesh Ranjan. Vlm-3r: Vision-language models augmented with instruction-aligned 3d reconstruction, 2025. URL <https://arxiv.org/abs/2505.20279>.
- [11] Huizhi Liang, Yichao Shen, Yu Deng, Sicheng Xu, Zhiyuan Feng, Tong Zhang, Yaobo Liang, and Jiaolong Yang. Hispatial: Taming hierarchical 3d spatial understanding in vision-language models, 2026. URL <https://arxiv.org/abs/2603.25411>.
- [12] Phillip Y. Lee, Jihyeon Je, Chanhoo Park, Mikaela Angelina Uy, Leonidas Guibas, and Minhyuk Sung. Perspective-aware reasoning in vision-language models via mental imagery simulation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. URL https://openaccess.thecvf.com/content/ICCV2025/html/Lee_Perspective-Aware_Reasoning_in_Vision-Language_Models_via_Mental_Impagery_Simulation_ICCV_2025_paper.html.